

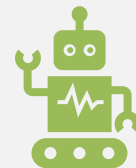
## TL;DR

- Goal: Predict a continuous target from anonymous numerical features.
- Main idea: Compare simple regularized linear models with more complex nonlinear models.
- Best observed submission: Private  $R^2 = 0.08499$  (XGBoost submission), outperforming Ridge on private leaderboard.
- Key lesson: Public leaderboard was noisy; private leaderboard favored better generalization.

# Competition Overview



Evaluation metric:  $R^2$ .

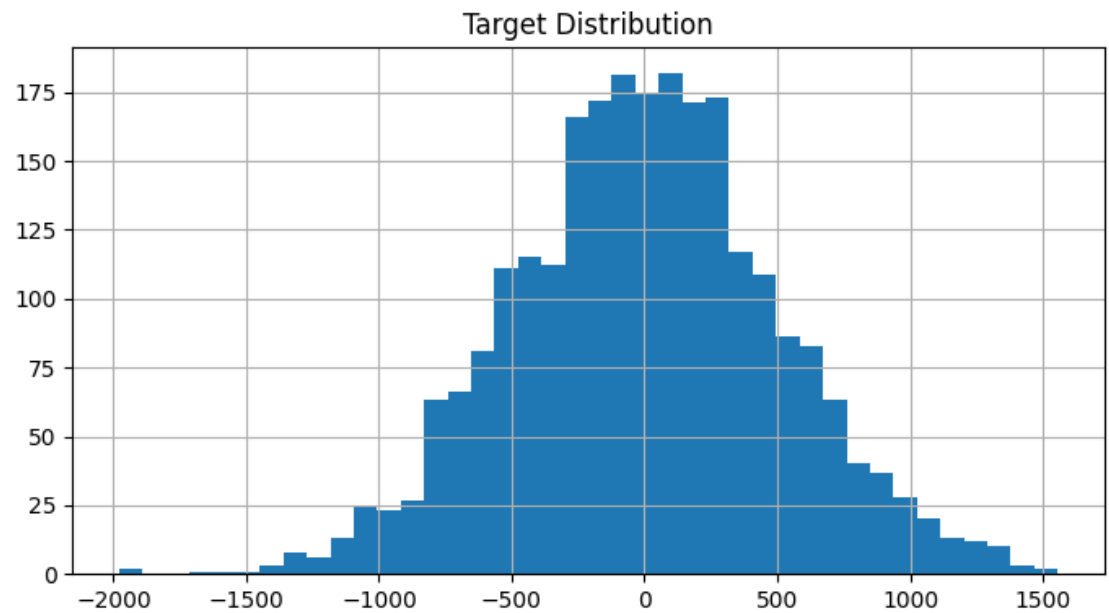


Anonymous features  
prevent domain-  
specific feature  
engineering.

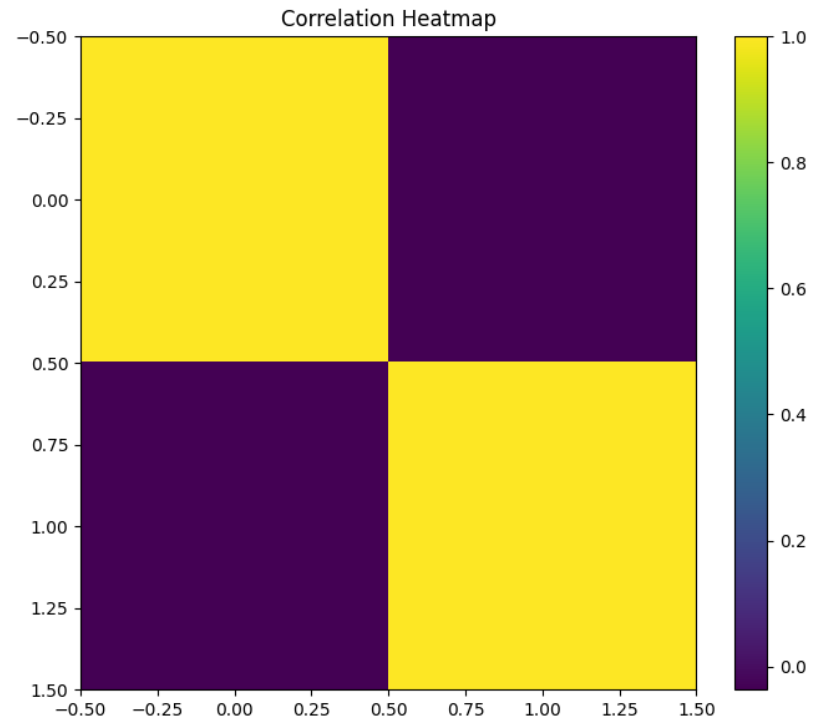


Focus on data  
quality, statistical  
relationships, and  
model robustness.

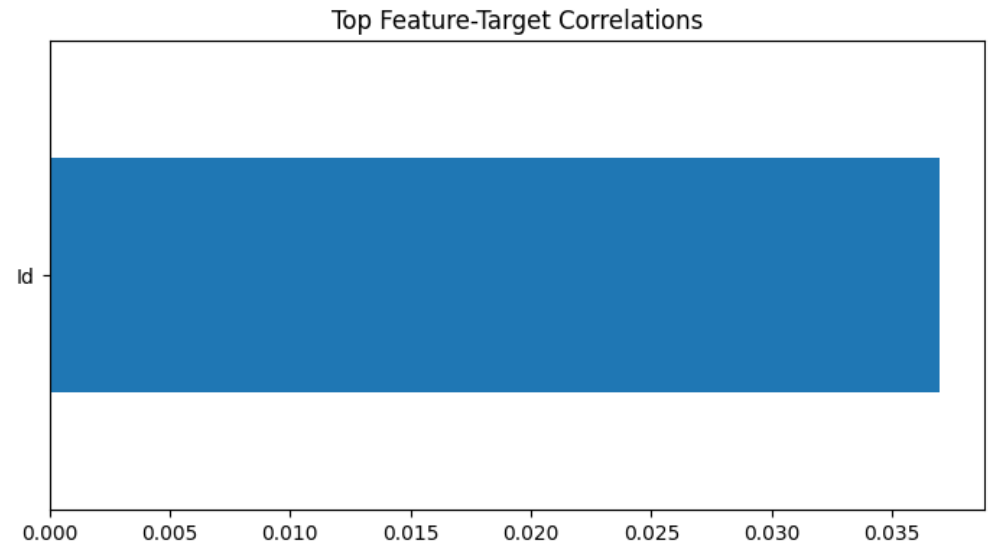
# EDA: Target Distribution



# EDA: Correlation Heatmap



# EDA: Top Feature-Target Correlations



# Experiment 1: Ridge Regression

Motivation: strong baseline, handles multicollinearity, robust to noisy features.

Pipeline: Median Imputation → StandardScaler → Ridge.

Result: Better than naive baseline but weak leaderboard performance.

Pros: Fast, interpretable, stable, less overfitting.

Cons: Linear only, no automatic interactions.

# Experiment 2: XGBoost / Gradient Boosting



Motivation: capture nonlinear relationships and feature interactions.



Expected advantage: anonymous features may contain hidden nonlinear signal.



Risk: overfitting due to small dataset and noisy target.



Private  $R^2$  improved from 0.06245 to 0.08499 (~36% improvement).

# Model Selection Strategy

Compare validation performance and leaderboard behavior.

Public leaderboard was unstable.

Private leaderboard used as primary indicator of generalization.



# Final Takeaways

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EDA revealed limited linear signal and missing values.

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Regularized linear models provided a stable benchmark.

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Boosting methods improved private leaderboard performance.

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Avoid optimizing exclusively for the public leaderboard.

# Code Repository



<https://www.kaggle.com/competitions/sofia-ml-regression-2026-spring/overview>